Sentence information load modulates comprehension of local versus global prediction, but not the N400 ERP

Comprehenders make predictions about upcoming content using previous information in the sentence, which contributes to language processing efficiency (Ryskin & Nieuwland, 2023). Capacity limitations on human cognitive resources mean that we may not always be able to keep all of the information in mind that might be relevant to making predictions (Futrell et al., 2020). The goal of the current study is to examine the impact of information load on sentence final prediction.

Think about these three sentences:

(B) "When the power went out, the house became…"

(S) "The winter weather was manageable with the heater on, but when the power went out, the house became…"

(L) "The winter weather was manageable with the heater on, but during the severe storm, when the power went out, the house became…"

Based on the cloze task, we know that the baseline sentence (B) leads to the prediction of "dark," but the shorter experimental sentence (S) leads to the prediction of "cold" due to the additional context provided in the first clause. The longer experimental sentence (L) involves an intervening middle clause, "during the severe storm", which does not provide information that alters the prediction, as confirmed by cloze data and LLM surprisal estimates. However, we hypothesized that this intervening clause would increase the information load and cognitive demand involved in maintaining that initial context that is critical in driving the prediction of "cold".

The current study examined whether the sentence information load and WM demand involved in processing the context impacted prediction, by using sentence contexts constructed like (S) versus (L). The key hypothesis was that the comprehenders will be more likely to generate a local prediction ("dark" in this example) based on the adjacent local context in the (L) sentences with the intervening clause.

The experiment used EEG recordings during a sentence comprehension task, with the key behavioral measure being comprehension judgments, and the key neural measure being the N400 event-related potential (ERP). Thirty native English speakers were recruited. Participants read sentences structured as (S) and (L), which were followed by a globally expected word ("cold"), a locally expected word ("dark"), or an implausible word ("bouncy"). Following each sentence, participants judged whether the sentence made sense. The N400 was measured following the sentence's final word.

Behavioral results showed a main effect of word ending, $F(2, 6024) = 1160.28$, $p < .001$. Participants responded to the globally expected word as highly sensible and to the implausible word ending as highly nonsensical, while the locally expected word fell in the middle. Importantly, there was a significant interaction between Ending × Length, $F(2, 6024) = 10.65$, $p < .001$. Participants responded that the local ending was more sensible, while the global ending was less sensible, following the (L) sentences than the (S) sentences (Figure 1). No difference was observed for the implausible ending. Such results support our hypothesis that participants' comprehension performance is impacted by the sentence information load, and they are more biased by the most adjacent context in the (L) sentence contexts with the intervening clause. The N400 ERP was analyzed using the mean amplitude values between 200 and 500 ms. Results showed the standard N400 effect following different endings, where greater prediction violations led to larger N400s (implausible > local > global), $F(2,58) = 31.51$, $p < .001$. However, no significant main effect of sentence length or significant interaction between Ending × Length was observed (Figure 2). The implications for the comprehension process that the N400 indexes will be discussed.

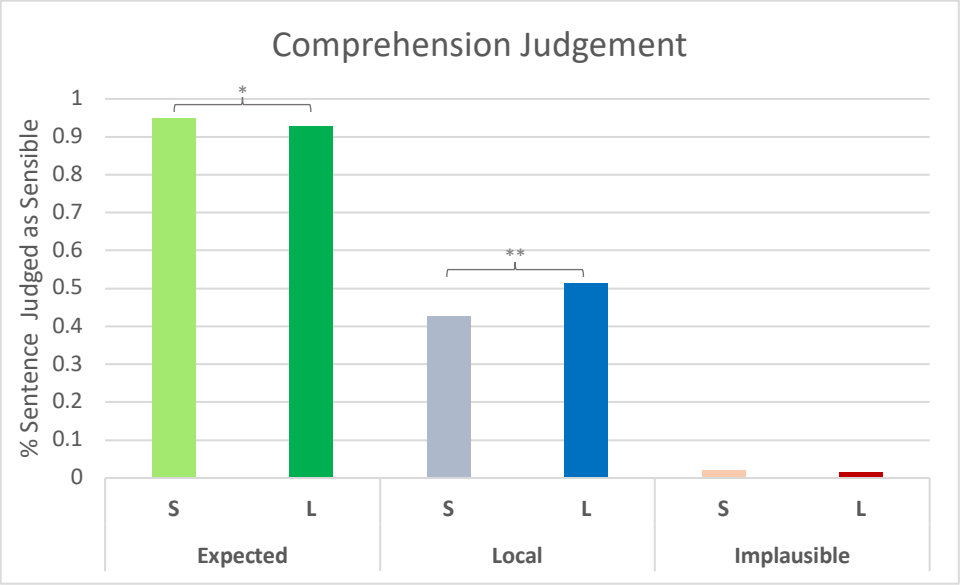

Figure 1. Behavioral judgment of the proportion of sentences that make sense

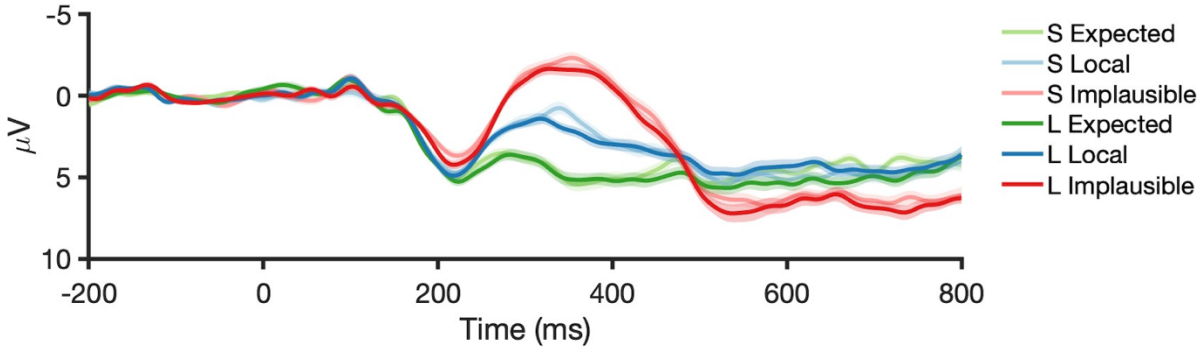

Figure 2. Averaged event-related potential waveforms are displayed for the short and long sentences, followed by the globally expected, local, or implausible word ending